

Pre-action state

State representation $z_i^{(r)} \in \mathbb{R}^{32}$



Retrieval progress



Answer & reasoning



Retrieved evidence



Information alignment

Which query should receive the next retrieval opportunity?

$\phi(\cdot)$

Queries

q_1
 q_2
 $\vdots$
 q_n



LLM

Answer + evidence

+
 +
 $\vdots$
 +



Single-step retrieval value

Pre-action state

$z_i^{(r)} \in \mathbb{R}^{32}$

$\dots$

What is the value of one more retrieval?

Value estimator

$f_\theta(\cdot)$



Estimated value

$\hat{\eta}_i^{(r)}$

train f_θ

Offline training supervision

$Q_i^{(r)}$

Retrieve once

$Q_i^{(r+1)}$

$$\Delta_i^{(r+1)} = Q_i^{(r+1)} - Q_i^{(r)}$$

Sequential shared-budget allocation

Shared budget B

$\dots$ (remaining)

1 budget unit

Current action pool

(query, action) $\hat{\eta}_i^{(r)}$

	$(q_7, a_7^{(1)})$	+0.08
	$(q_4, a_4^{(1)})$	+0.06
	$(q_9, a_9^{(2)})$	+0.04
	$\vdots$	
	$(q_2, a_2^{(1)})$	-0.01

Next allocation step

Updated action pool

(query, action) $\hat{\eta}_i^{(r)}$

	$(q_4, a_4^{(1)})$	+0.06
	$(q_9, a_9^{(2)})$	+0.04
	$(q_7, a_7^{(2)})$	+0.03
	$\vdots$	
	$(q_2, a_2^{(1)})$	-0.01

One allocation step

Retrieve once

Update state

$z^{(r)} \rightarrow z^{(r+1)}$

Next legal action?

Yes

Re-estimate

$f_\theta(\cdot)$

Reinsert



No

Exit pool